AI Assisted Coding
Assignment Lab - 17

Name : Madarapu Likhith Sai
Hall.t.No. : 2303A51026
Batch No. : 01

Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts

Lab Objectives:

- Learn how to clean raw datasets using AI-assisted Python scripting.
- Apply preprocessing techniques such as handling missing values, encoding categorical data, and normalization.
- Automate repetitive data-cleaning tasks with AI-generated code.
- Understand how preprocessing impacts model performance.

Task 1: Perform EDA on Penguin dataset(from seaborn)

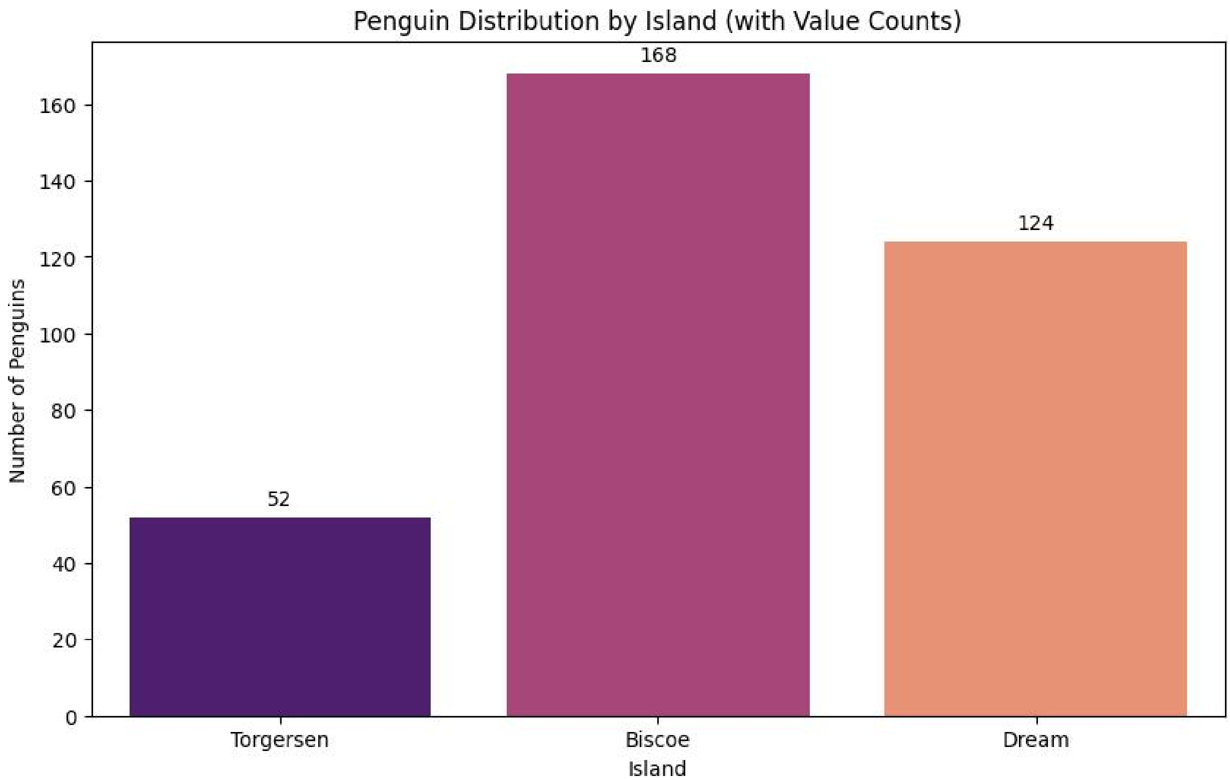
- Load and display entire dataset
- Display country wise count
- Visualize bar chart (country wise)
- Visualize bar chart with values count at top
- Visualize pairplot and other plots

Output :

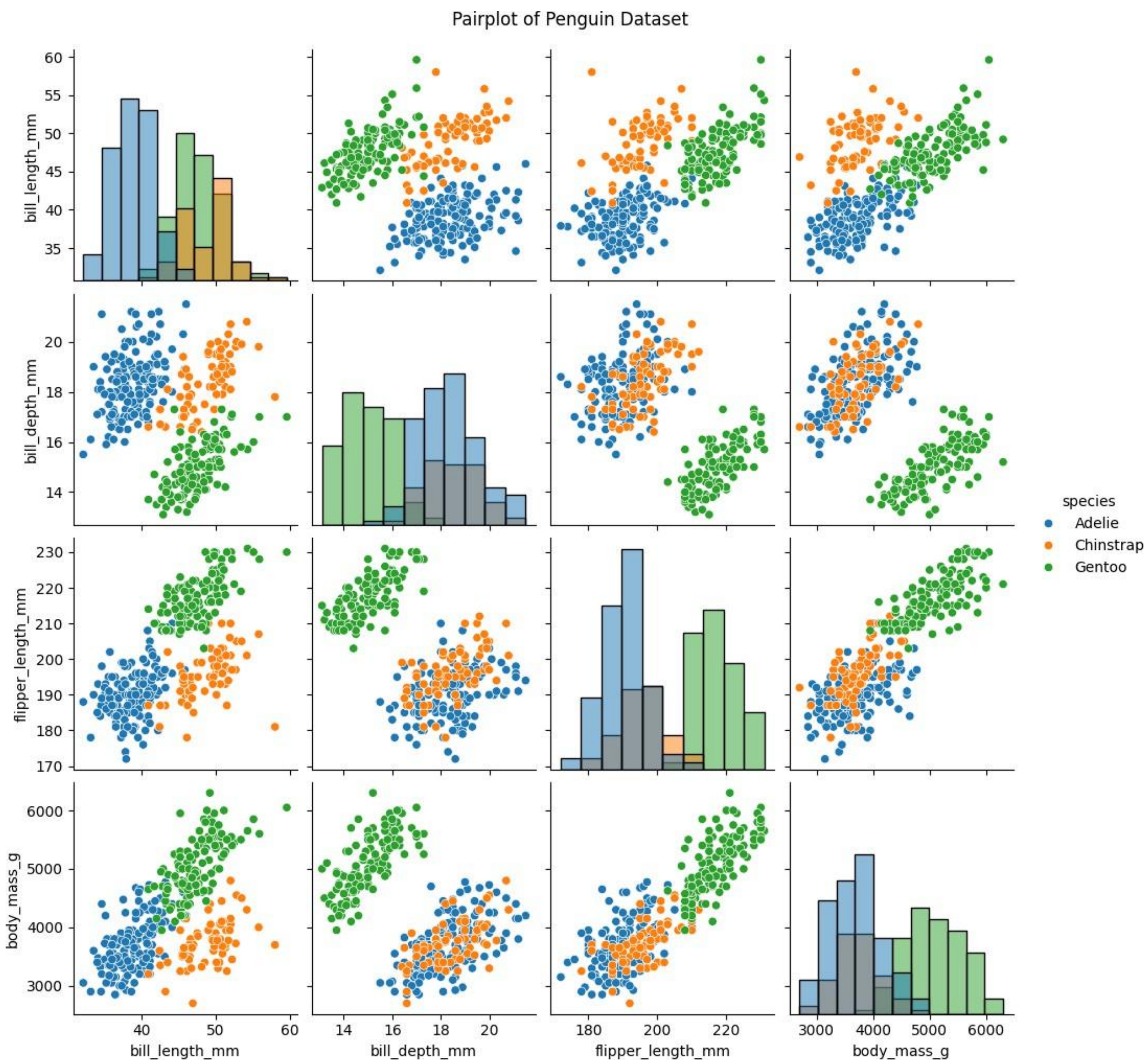
320	Gentoo	Biscoe	48.5	15.0	219.0	4850.0	Female
321	Gentoo	Biscoe	55.9	17.0	228.0	5600.0	Male
322	Gentoo	Biscoe	47.2	15.5	215.0	4975.0	Female
323	Gentoo	Biscoe	49.1	15.0	228.0	5500.0	Male
324	Gentoo	Biscoe	47.3	13.8	216.0	4725.0	NaN
325	Gentoo	Biscoe	46.8	16.1	215.0	5500.0	Male
326	Gentoo	Biscoe	41.7	14.7	210.0	4700.0	Female
327	Gentoo	Biscoe	53.4	15.8	219.0	5500.0	Male
328	Gentoo	Biscoe	43.3	14.0	208.0	4575.0	Female
329	Gentoo	Biscoe	48.1	15.1	209.0	5500.0	Male
330	Gentoo	Biscoe	50.5	15.2	216.0	5000.0	Female
331	Gentoo	Biscoe	49.8	15.9	229.0	5950.0	Male
332	Gentoo	Biscoe	43.5	15.2	213.0	4650.0	Female
333	Gentoo	Biscoe	51.5	16.3	230.0	5500.0	Male
334	Gentoo	Biscoe	46.2	14.1	217.0	4375.0	Female
335	Gentoo	Biscoe	55.1	16.0	230.0	5850.0	Male
336	Gentoo	Biscoe	44.5	15.7	217.0	4875.0	NaN
337	Gentoo	Biscoe	48.8	16.2	222.0	6000.0	Male
338	Gentoo	Biscoe	47.2	13.7	214.0	4925.0	Female
339	Gentoo	Biscoe	NaN	NaN	NaN	NaN	NaN
340	Gentoo	Biscoe	46.8	14.3	215.0	4850.0	Female
341	Gentoo	Biscoe	50.4	15.7	222.0	5750.0	Male
342	Gentoo	Biscoe	45.2	14.8	212.0	5200.0	Female
343	Gentoo	Biscoe	49.9	16.1	213.0	5400.0	Male

Country/Island wise count:

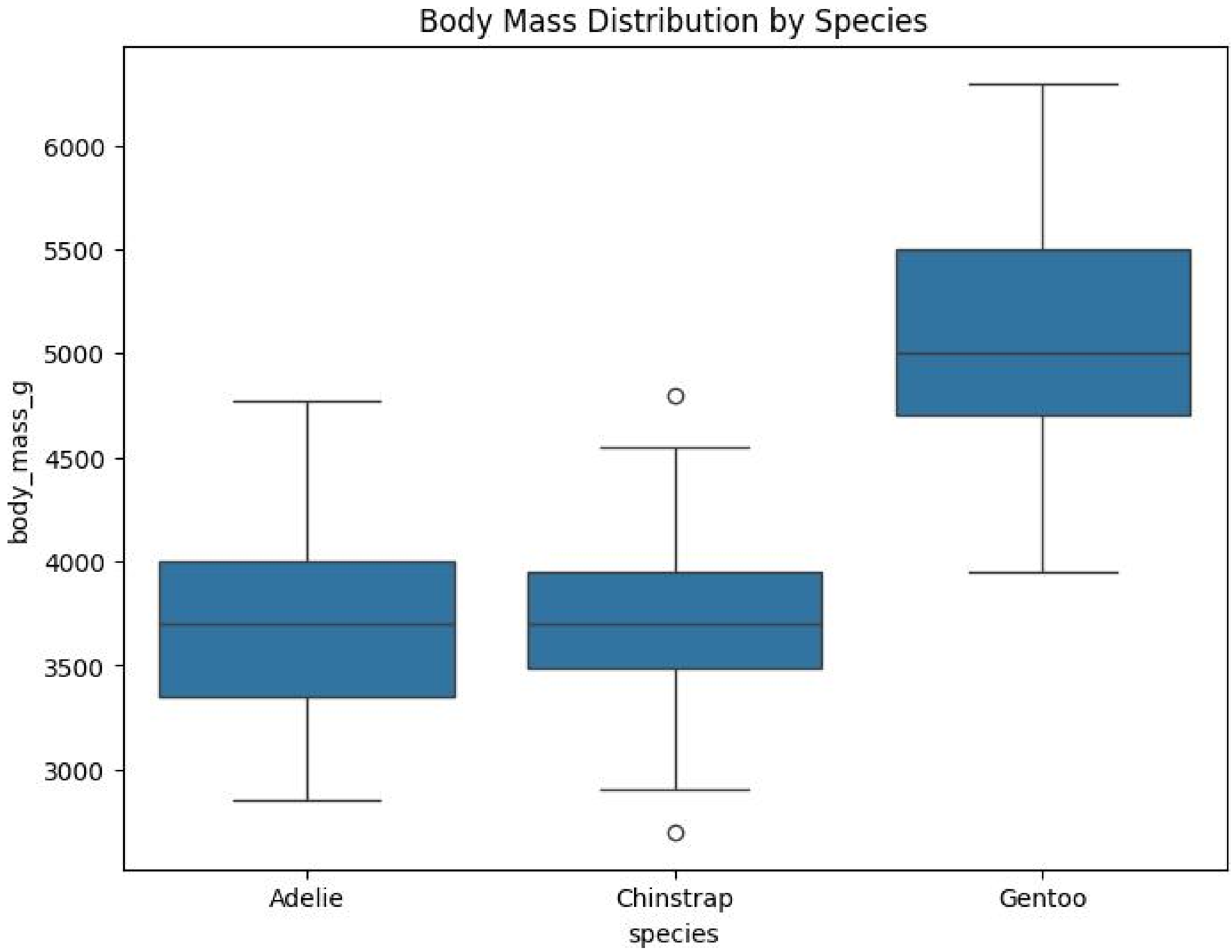
	island	count
0	Biscoe	168
1	Dream	124
2	Torgersen	52



Penguin Distribution by Island (with Value Counts)
168
124
52
Number of Penguins
0
20
40
60
80
100
120
140
160
Torgersen
Biscoe
Dream
Island



Pairplot of Penguin Dataset
bill_length_mm
bill_depth_mm
flipper_length_mm
body_mass_g
species
Adelie
Chinstrap
Gentoo



Task 2: Perform EDA on HealthExp dataset(seaborn)

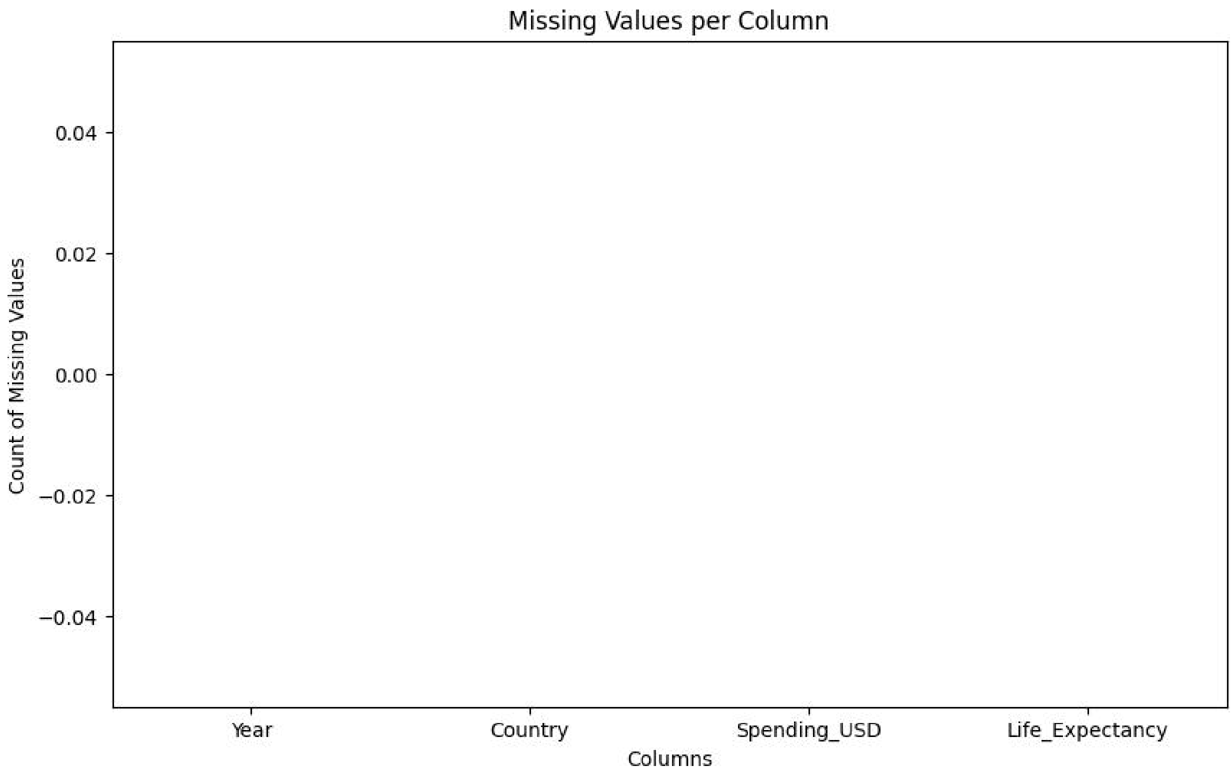
- Load and display entire dataset
- Display columns and statistical summary (describe)
- Display missing values , and draw bar chart
- Visualize pairplot, regplot and box plot, scatter plot etc

250	2017	Canada	5150.470	81.9
251	2017	Germany	5970.163	81.1
252	2017	France	5005.756	82.7
253	2017	Great Britain	4059.125	81.3
254	2017	Japan	4412.852	84.2
255	2017	USA	10046.472	78.6
256	2018	Canada	5308.356	82.0
257	2018	Germany	6281.840	81.0
258	2018	France	5099.306	82.8
259	2018	Great Britain	4189.708	81.3
260	2018	Japan	4554.276	84.3
261	2018	USA	10451.386	78.7
262	2019	Canada	5189.721	82.2
263	2019	Germany	6407.928	81.3
264	2019	France	5167.839	82.9
265	2019	Great Britain	4385.463	81.4
266	2019	Japan	4610.794	84.4
267	2019	USA	10855.517	78.8
268	2020	Canada	5828.324	81.7
269	2020	Germany	6938.983	81.1
270	2020	France	5468.418	82.3
271	2020	Great Britain	5018.700	80.4
272	2020	Japan	4665.641	84.7
273	2020	USA	11859.179	77.0

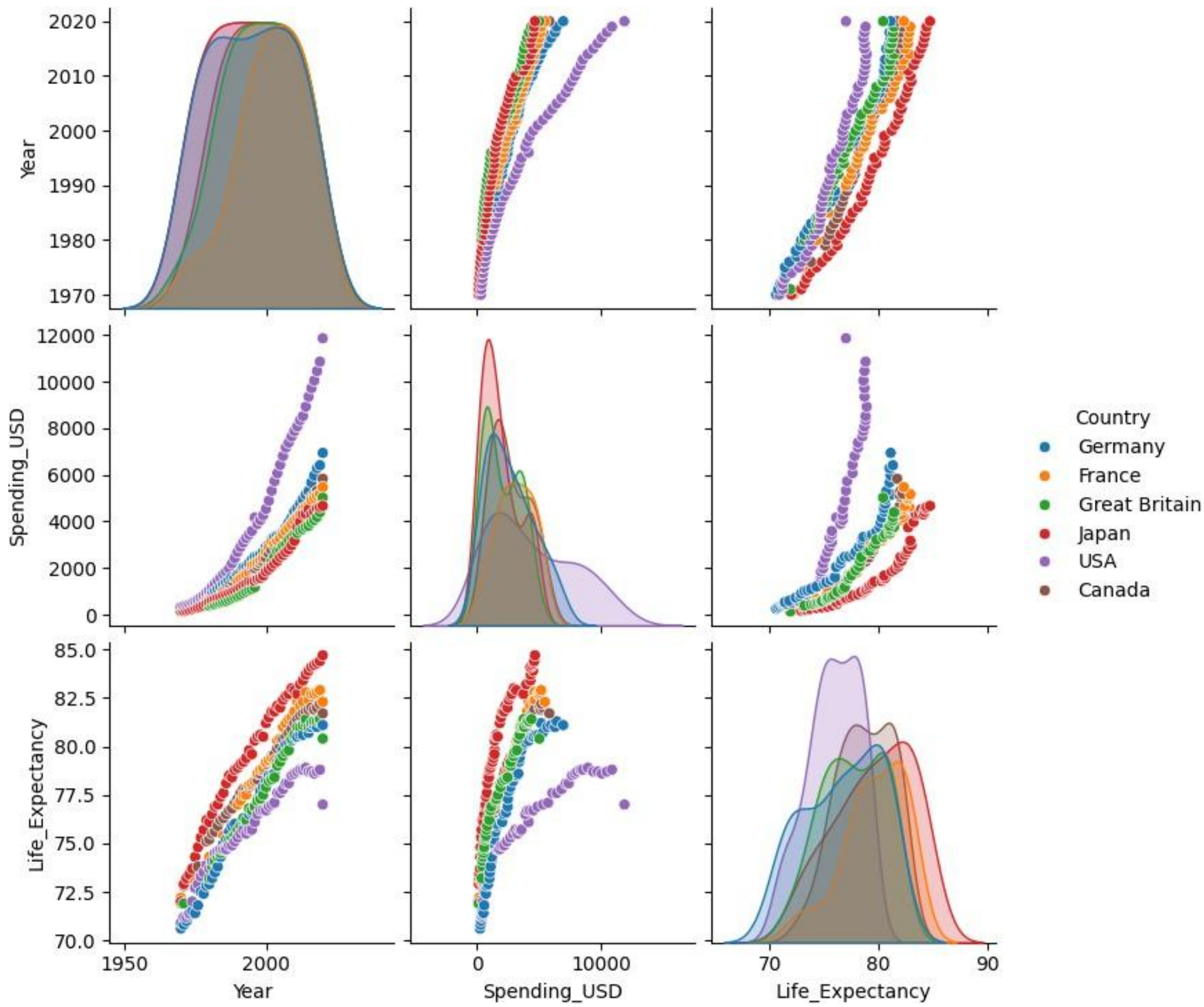
```
Columns: ['Year', 'Country', 'Spending_USD', 'Life_Expectancy']

Statistical Summary:
```

	Year	Country	Spending_USD	Life_Expectancy
count	274.000000	274	274.000000	274.000000
unique	NaN	6	NaN	NaN
top	NaN	USA	NaN	NaN
freq	NaN	51	NaN	NaN
mean	1996.992701	NaN	2789.338905	77.909489
std	14.180933	NaN	2194.939785	3.276263
min	1970.000000	NaN	123.993000	70.600000
25%	1985.250000	NaN	1038.357000	75.525000
50%	1998.000000	NaN	2295.578000	78.100000
75%	2009.000000	NaN	4055.610000	80.575000
max	2020.000000	NaN	11859.179000	84.700000

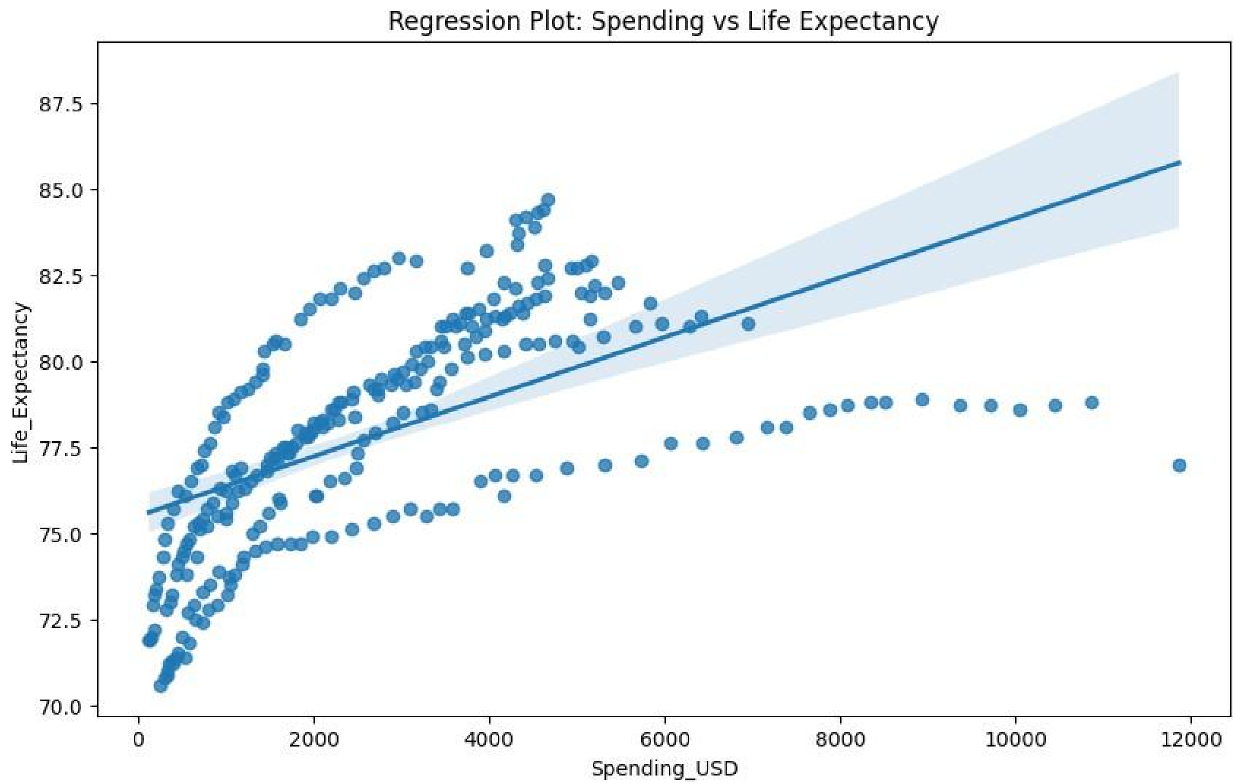
```
Missing Values per Column:
Year               0
Country            0
Spending_USD       0
Life_Expectancy    0
dtype: int64
```



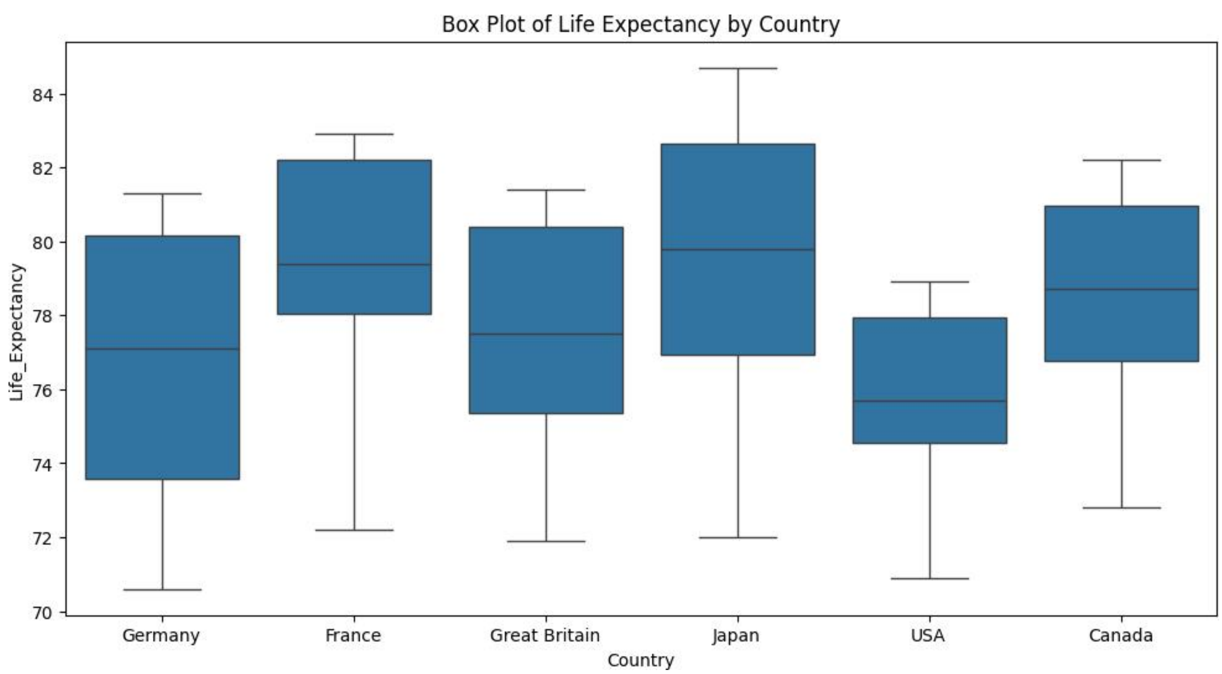
Missing Values per Column
Count of Missing Values
0.04
0.02
0.00
−0.02
−0.04
Year
Country
Spending_USD
Life_Expectancy
Columns



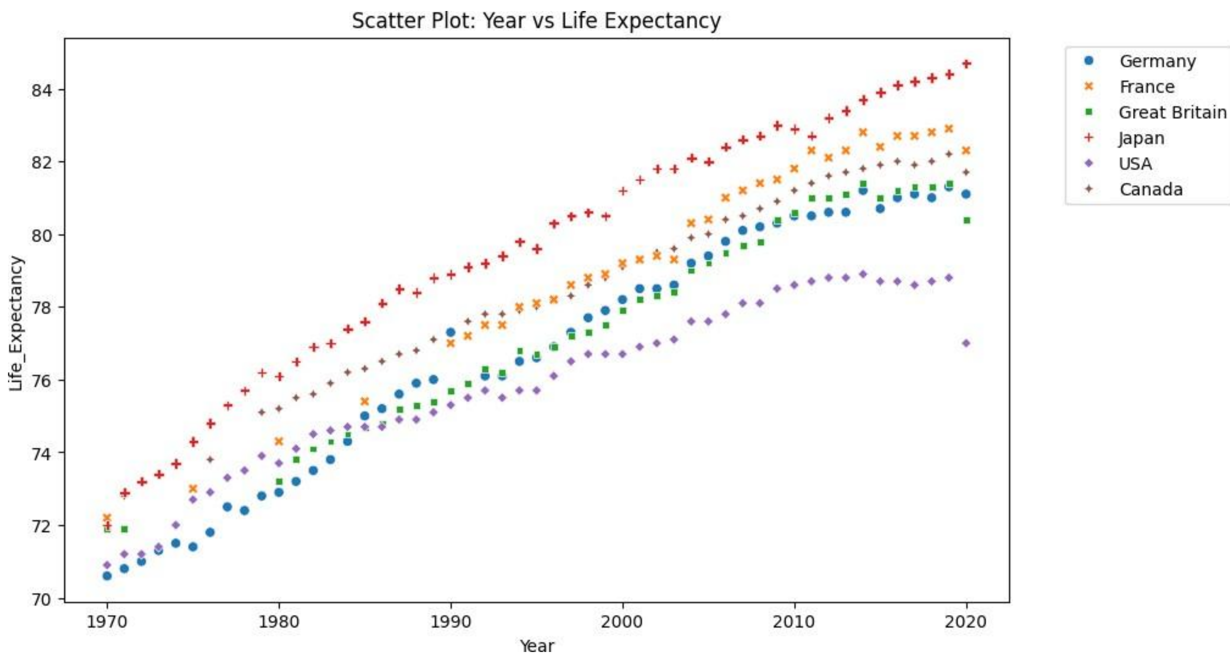
Year
Spending_USD
Life_Expectancy
Country
Germany
France
Great Britain
Japan
USA
Canada



Regression Plot: Spending vs Life Expectancy
Life_Expectancy
Spending_USD
87.5
85.0
82.5
80.0
77.5
75.0
72.5
70.0
0
2000
4000
6000
8000
10000
12000



Box Plot of Life Expectancy by Country
Life_Expectancy
Country
84
82
80
78
76
74
72
70
Germany
France
Great Britain
Japan
USA
Canada



Task 3: Perform following task on given dataset (Train the model)

- Analyze the Planets dataset to understand its structure, features, and missing values.
- Perform data cleaning and preprocessing to handle missing or incorrect data.
- Train the dataset using different machine learning models to make predictions.
- Compare and explain the performance of all trained models with proper reasoning.

```
Dataset Shape: (1035, 6)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1035 entries, 0 to 1034
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   method          1035 non-null   object 
 1   number          1035 non-null   int64  
 2   orbital_period  992 non-null    float64
 3   mass            513 non-null    float64
 4   distance        808 non-null    float64
 5   year            1035 non-null   int64  
dtypes: float64(3), int64(2), object(1)
memory usage: 48.6+ KB
None
```

	number	orbital_period	mass	distance	year
count	1035.000000	992.000000	513.000000	808.000000	1035.000000
mean	1.785507	2002.917596	2.638161	264.069282	2009.070531
std	1.240976	26014.728304	3.818617	733.116493	3.972567
min	1.000000	0.090706	0.003600	1.350000	1989.000000
25%	1.000000	5.442540	0.229000	32.560000	2007.000000
50%	1.000000	39.979500	1.260000	55.250000	2010.000000
75%	2.000000	526.005000	3.040000	178.500000	2012.000000
max	7.000000	730000.000000	25.000000	8500.000000	2014.000000

	method	number	orbital_period	mass	distance	year
0	Radial Velocity	1	269.300	7.10	77.40	2006
1	Radial Velocity	1	874.774	2.21	56.95	2008
2	Radial Velocity	1	763.000	2.60	19.84	2011
3	Radial Velocity	1	326.030	19.40	110.62	2007
4	Radial Velocity	1	516.220	10.50	119.47	2009

```
Missing values per column:
 method             0
number             0
orbital_period    43
mass             522
distance         227
year               0
dtype: int64

Cleaned Dataset Shape: (498, 6)
```

	Model	MSE	R2 Score
0	Linear Regression	3.474183e+06	0.056249
1	Random Forest	3.524382e+06	0.042612
2	Gradient Boosting	3.107439e+06	0.155874

- **Linear Regression**: Acts as a baseline. It might struggle if the relationships between features (like mass/distance) and the orbital period are non-linear.
- **Random Forest/Gradient Boosting**: These ensemble methods usually perform better as they capture non-linear interactions and are more robust to outliers in the astronomical data.
- **Performance**: The model with the highest R2 score and lowest MSE is the best performer for this specific subset of data.

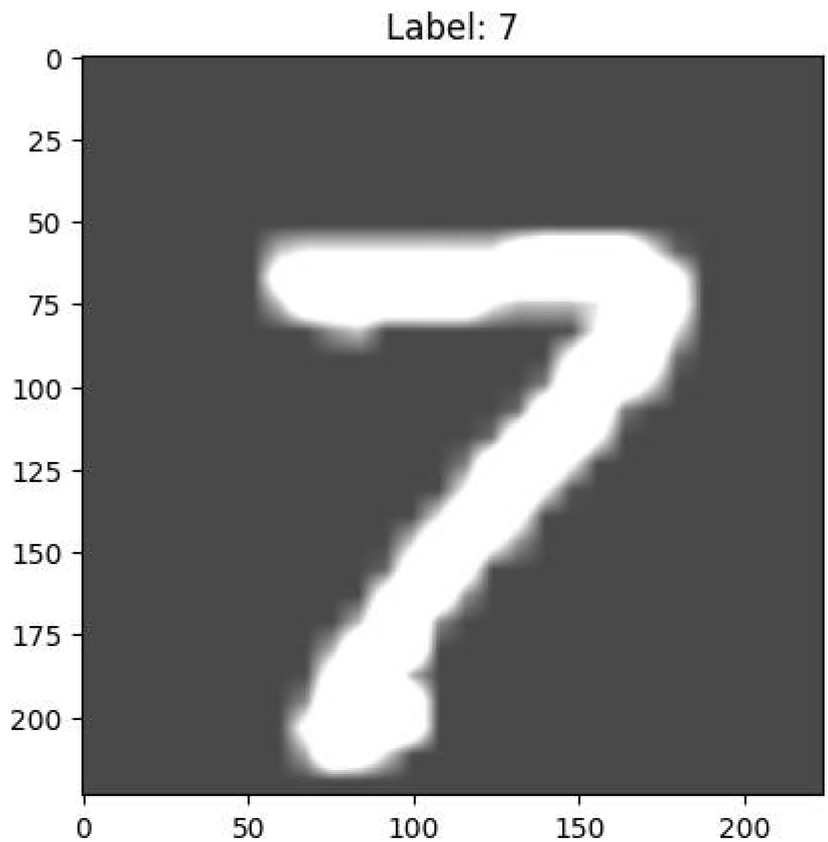
Task 4: Perform following task on given dataset (Train the model)

- Load the MNIST dataset from PyTorch.
- Display basic information and visualize sample input images with their class labels.
- Perform data preprocessing
- Design a EfficientNet model for image classification.
- Train the CNN model (10 epochs)
- Evaluate model performance using accuracy and loss metrics.
- Visualize the training history (accuracy and loss curves).

```
Using device: cuda
```

```
100%|██████████| 9.91M/9.91M [00:00<00:00, 19.6MB/s]
100%|██████████| 28.9k/28.9k [00:00<00:00, 484kB/s]
100%|██████████| 1.65M/1.65M [00:00<00:00, 4.46MB/s]
100%|██████████| 4.54k/4.54k [00:00<00:00, 12.9MB/s]Training samples: 60000
Testing samples: 10000
```

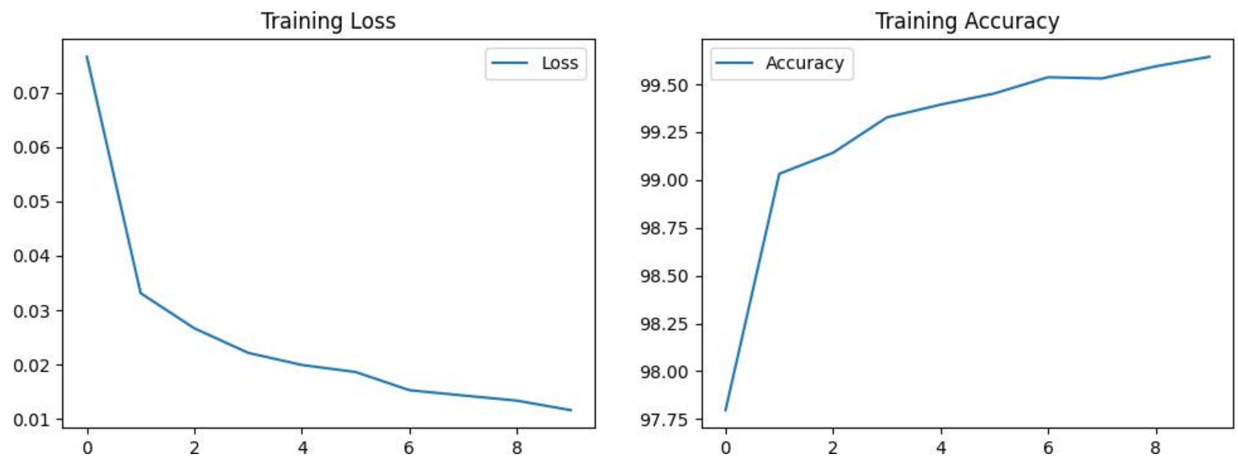
```
WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers). Got range [0.28789353..1.9107434].
```




```
Downloading: "https://download.pytorch.org/models/efficientnet_b0_rwightman-7f5810bc.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_b0_rwightman-7f5810bc.pth
100%|██████████| 20.5M/20.5M [00:00<00:00, 119MB/s]
```

```
Epoch 1/10 - Loss: 0.0766, Acc: 97.80%
Epoch 2/10 - Loss: 0.0331, Acc: 99.03%
Epoch 3/10 - Loss: 0.0266, Acc: 99.14%
Epoch 4/10 - Loss: 0.0221, Acc: 99.33%
Epoch 5/10 - Loss: 0.0199, Acc: 99.39%
Epoch 6/10 - Loss: 0.0186, Acc: 99.45%
Epoch 7/10 - Loss: 0.0153, Acc: 99.54%
Epoch 8/10 - Loss: 0.0143, Acc: 99.53%
Epoch 9/10 - Loss: 0.0133, Acc: 99.59%
Epoch 10/10 - Loss: 0.0116, Acc: 99.64%
```

```
Final Test Accuracy: 99.46%
```



Task 5 – Customer Data Cleaning

Task: Use AI to generate a Python script for cleaning a customer dataset.

Instructions:

- Handle missing values in columns (age, income, city).
- Remove duplicate records.
- Standardize text values (e.g., city names in lowercase).
- Encode categorical variables (gender, city).

Sample Code:

import pandas as pd

data = pd.read_csv("shopping_trends.csv")

print(data.head())

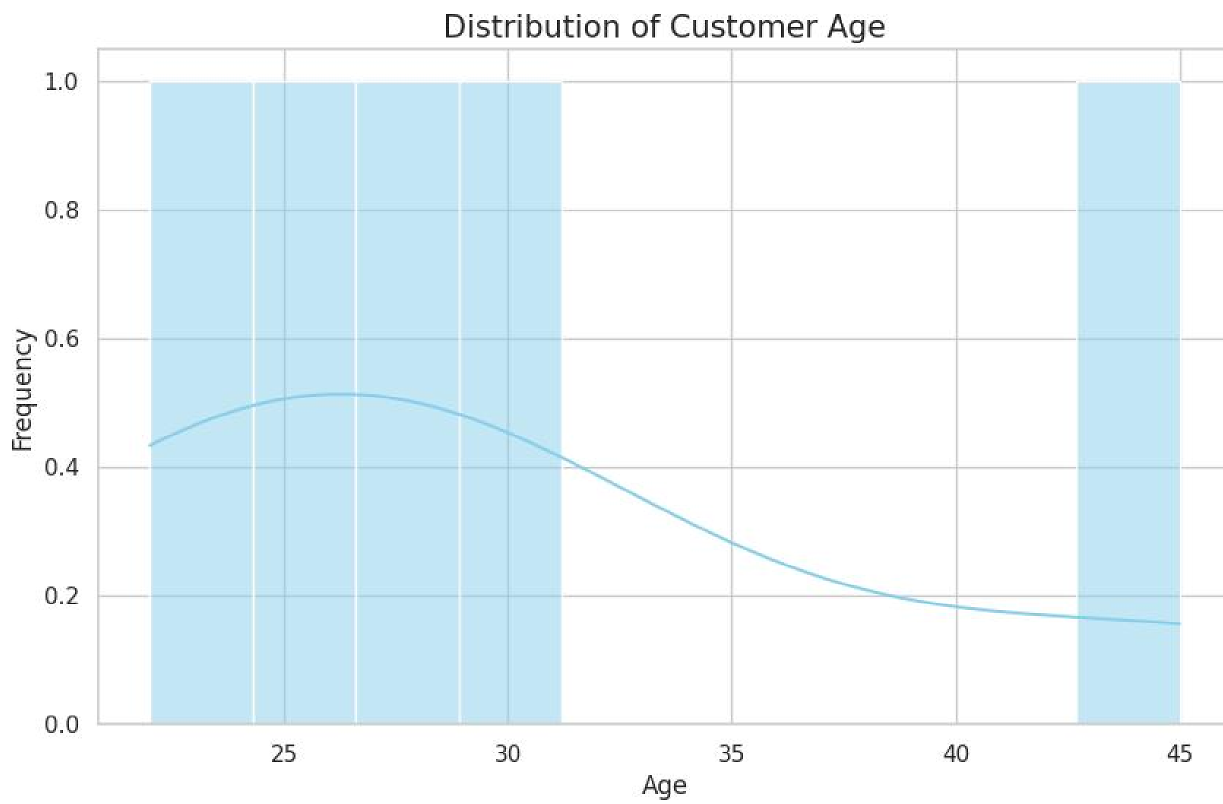
Dataset link:
https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset?select=shopping_trends.csv
Expected Output:

- A cleaned Pandas data frame ready for analysis.

	Customer ID	Age	Income	City	Gender
0	1	25.0	50000.0	New York	Male
1	2	NaN	60000.0	Los Angeles	Female
2	2	30.0	60000.0	Los Angeles	Female
3	3	45.0	NaN	new york	Male
4	4	22.0	55000.0	Chicago	Female

```
Cleaning complete.
```

	Customer ID	Age	Income	City	Gender	Gender_Encoded	City_Encoded
0	1	25.0	50000.0	new york	Male	1	2
1	2	27.5	60000.0	los angeles	Female	0	1
2	2	30.0	60000.0	los angeles	Female	0	1
3	3	45.0	56250.0	new york	Male	1	2
4	4	22.0	55000.0	chicago	Female	0	0



Task 6 – Sales Data Preprocessing

Task: Preprocess sales transaction data using AI assistance.

Instructions:

- Convert date columns into datetime format.
- Create new features (month, year, day of week).
- Normalize sales amount column.
- Detect and handle outliers in dataset.

Sample Code:

```
import pandas as pd
data = pd.read_csv("sales_data.csv")
print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/vinothkannaece/sales-dataset?select=sales_data.csv

Expected Output:

• A transformed dataset with time-based features and normalized values.

```
Original Data:
```

	date	sales_amount
0	2023-01-01	100
1	2023-02-15	150
2	2023-03-10	2000
3	2023-04-05	120
4	2023-12-31	180

	date	sales_amount	year	month	day_of_week
0	2023-01-01	100	2023	1	Sunday
1	2023-02-15	150	2023	2	Wednesday
2	2023-03-10	2000	2023	3	Friday
3	2023-04-05	120	2023	4	Wednesday
4	2023-12-31	180	2023	12	Sunday

	sales_amount	normalized_sales
0	100	0.000000
1	150	0.010204
2	2000	0.387755
3	120	0.004082
4	180	0.016327

```
Original rows: 6
Rows after outlier removal: 5
```

	date	sales_amount	year	month	day_of_week	normalized_sales
0	2023-01-01	100	2023	1	Sunday	0.000000
1	2023-02-15	150	2023	2	Wednesday	0.010204
2	2023-03-10	2000	2023	3	Friday	0.387755
3	2023-04-05	120	2023	4	Wednesday	0.004082
4	2023-12-31	180	2023	12	Sunday	0.016327

Task 7 – Healthcare Data Preparation

Task: Clean and preprocess a healthcare dataset using AI scripts.

Instructions:

- Handle missing values in blood_pressure, cholesterol.
- Encode categorical features
- Scale numerical features (min-max or standard scaling).
- Split dataset into training and testing sets.

Dataset link :

https://www.kaggle.com/datasets/abdallaahmed77/healthcare-risk-factors-dataset

Expected Output:

- A preprocessed dataset ready for machine learning models.

```
Dataset loaded successfully.
Filled missing blood_pressure with median: 130.0
Filled missing cholesterol with median: 200.0
Encoded categorical columns: ['gender', 'smoking_status']
Numerical features scaled.

Preprocessing complete.
Training shape: (80, 5), Testing shape: (20, 5)
```

	age	gender	blood_pressure	cholesterol	smoking_status	risk_score
0	0.305313	1.105542	0.094811	-0.862662	0.904534	1
1	-1.295357	1.105542	0.094811	-0.323498	-1.105542	0
2	1.550278	-0.904534	-1.259635	-0.323498	0.904534	1
3	0.779585	1.105542	-1.259635	-0.323498	-1.105542	0
4	-0.524664	-0.904534	1.449258	-0.323498	0.904534	1

```
Missing values handled.
```

```
Encoded categorical columns: []
```

```
Numerical features scaled using StandardScaler.
```

	age	gender	blood_pressure	cholesterol	smoking_status	risk_score
0	0.305313	1.105542	0.094811	-0.862662	0.904534	1
1	-1.295357	1.105542	0.094811	-0.323498	-1.105542	0
2	1.550278	-0.904534	-1.259635	-0.323498	0.904534	1
3	0.779585	1.105542	-1.259635	-0.323498	-1.105542	0
4	-0.524664	-0.904534	1.449258	-0.323498	0.904534	1

```
Data split completed.
Training set shape: (80, 5)
Testing set shape: (20, 5)
```

Task 8 – Employee Salary Data Preprocessing

Task: Clean and preprocess an employee salary dataset using AI scripts.

Instructions:

- Handle missing values
- Detect and remove salary outliers.

• Standardize department names (lowercase).
• Apply label encoding to job location, department.

Dataset link:

https://www.kaggle.com/code/vishantmathur/employee-salary

Expected Output:

• A structured dataset ready for HR analytics.

```
Initial Data Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 3 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   salary        9 non-null      float64
 1   department    9 non-null      object
 2   job_location  10 non-null     object
dtypes: float64(1), object(2)
memory usage: 372.0+ bytes
None
```

	salary	department	job_location
0	50000.0	HR	New York
1	60000.0	Engineering	Remote
2	1200000.0	hr	New York
3	55000.0	MARKETING	Chicago
4	NaN	Engineering	Remote

```
Missing values after cleaning:
salary          0
department      0
job_location    0
dtype: int64
```

```
Standardized Departments:
['hr' 'engineering' 'marketing' 'sales']
```

```
Rows before outlier removal: 10
Rows after outlier removal: 9
```

Final Preprocessed Dataset:

	salary	department	job_location	job_location_encoded	department_encoded
0	50000.0	hr	New York	1	1
1	60000.0	engineering	Remote	2	0
3	55000.0	marketing	Chicago	0	2
4	58000.0	engineering	Remote	2	0
5	58000.0	sales	Chicago	0	3

Data ready for HR Analytics.

Task 9 –Loan Approval Dataset Cleaning

Use AI to generate preprocessing code for a loan dataset. Instructions:

- Handle missing values in loan_amount, credit_score.
- Encode loan_status (Approved/Rejected).
- Scale numerical features using standard scaling.
- Split the dataset into training and testing sets.

Sample Code:

```
import pandas as pd
data = pd.read_csv("loadapproval.csv")
print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/amineipad/loan-approval-dataset

Expected Output:

- A machine-learning-ready loan dataset.

```
Sample Data Preview (Using generated data for demonstration):
```

	loan_amount	credit_score	income	loan_status
0	100.0	700.0	50000	Approved
1	200.0	650.0	60000	Rejected
2	NaN	710.0	55000	Approved
3	300.0	NaN	80000	Approved
4	150.0	600.0	40000	Rejected

```
Encoded classes: ['Approved', 'Rejected']

Data after Cleaning and Scaling:
```

	loan_amount	credit_score	income	loan_status
0	-1.279754	0.092848	-0.767983	0
1	-0.389490	-0.835629	-0.335316	1
2	-0.166924	0.278543	-0.551650	0
3	0.500773	0.092848	0.530016	0
4	-0.834622	-1.764106	-1.200649	1

```
Training set shape: (6, 3)
Testing set shape: (2, 3)

Preprocessing Complete: The dataset is now machine-learning-ready.
```

Task 10 – Social Media Profiles Preprocess a social media engagement dataset using AI assistance.

Instructions:

- Remove duplicate user profiles.
- Handle missing values

- Encode categorical columns

Dataset link:
https://www.kaggle.com/datasets/medaxone/top-100-social-media-profiles

Expected Output:

- A cleaned dataset for engagement prediction.

```
Preprocessing function for Social Media Profiles defined.
```

```
--- Original Data Sample ---
```

	Username	Category	Followers	Engagement_Rate
0	cristiano	Sports	600.0	2.5
1	leomessi	Sports	480.0	3.1
2	selenagomez	Music	430.0	NaN
3	cristiano	Sports	600.0	2.5
4	therock	Entertainment	390.0	1.8
5	arianagrande	None	380.0	2.2
6	None	Music	NaN	2.0

```
Removed 1 duplicate profiles.

--- Cleaned Dataset Summary ---
<class 'pandas.core.frame.DataFrame'>
Index: 6 entries, 0 to 6
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Username           6 non-null      object
 1   Category           6 non-null      object
 2   Followers          6 non-null      float64
 3   Engagement_Rate    6 non-null      float64
 4   Category_encoded   6 non-null      int64
dtypes: float64(2), int64(1), object(2)
memory usage: 288.0+ bytes
None
```

--- Statistics of Cleaned Data ---

	Username	Category	Followers	Engagement_Rate	Category_encoded
count	6	6	6.000000	6.00000	6.000000
unique	5	3	NaN	NaN	NaN
top	arianagrande	Music	NaN	NaN	NaN
freq	2	3	NaN	NaN	NaN
mean	NaN	NaN	451.666667	2.30000	1.166667
std	NaN	NaN	80.849655	0.45607	0.752773
min	NaN	NaN	380.000000	1.80000	0.000000
25%	NaN	NaN	400.000000	2.05000	1.000000
50%	NaN	NaN	430.000000	2.20000	1.000000
75%	NaN	NaN	467.500000	2.42500	1.750000
max	NaN	NaN	600.000000	3.10000	2.000000

--- Final Cleaned DataFrame ---

	Username	Category	Followers	Engagement_Rate	Category_encoded
0	cristiano	Sports	600.0	2.5	2
1	leomessi	Sports	480.0	3.1	2
2	selenagomez	Music	430.0	2.2	1
4	therock	Entertainment	390.0	1.8	0
5	arianagrande	Music	380.0	2.2	1
6	arianagrande	Music	430.0	2.0	1

Task 11 – Weather Dataset Feature Engineering

Task: Use AI to preprocess a weather dataset for forecasting.

Instructions:

- Convert date column into datetime format.
- Handle missing values in temperature, humidity.
- Create new features (season, week_number).

• Normalize rainfall values.

Dataset link:

https://www.kaggle.com/code/alsamaualalhinai/weather-dataset/input

Expected Output:

• A processed dataset ready for forecasting models.

```
File not found. Generating dummy data for demonstration...
```

	Date	Temperature	Humidity	Rainfall
0	2022-01-01	32.259124	79.866835	0.540408
1	2022-01-02	31.361620	48.231455	1.686527
2	2022-01-03	24.874405	82.685214	2.408219
3	2022-01-04	25.745459	73.616389	2.442471
4	2022-01-05	16.707279	85.994746	0.258878

```
Processing complete.
```

	Date	Temperature	Humidity	Rainfall	week_number	season	Rainfall_normalized
0	2022-01-01	32.259124	79.866835	0.540408	52	Winter	0.053692
1	2022-01-02	31.361620	48.231455	1.686527	52	Winter	0.170264
2	2022-01-03	24.874405	82.685214	2.408219	1	Winter	0.243667
3	2022-01-04	25.745459	73.616389	2.442471	1	Winter	0.247151
4	2022-01-05	16.707279	85.994746	0.258878	1	Winter	0.025058

Task 12 – Movie Ratings Dataset Preparation

Task: Clean and preprocess a movie ratings dataset using AI- generated scripts.

Instructions:

• Handle missing values in rating, genre.

• Encode categorical variables like genre, language.

• Remove duplicate movie entries.

• Scale rating values between 0 and 1.

Dataset link:

https://www.kaggle.com/datasets/PromptCloudHQ/imdb-data

Expected Output:

• A dataset suitable for recommendation systems.

	Title	Genre	Rating	Revenue (Millions)	Metascore
0	Inception	Action,Adventure,Sci-Fi	8.8	292.57	74.0
1	Interstellar	Adventure,Drama,Sci-Fi	8.6	187.99	74.0
2	The Dark Knight	Action,Crime,Drama	9.0	533.32	84.0
3	Inception	Action,Adventure,Sci-Fi	8.8	292.57	74.0
4	Dunkirk	Action,Drama,History	7.9	188.37	94.0

Preprocessing Complete!

	Title	Rating_Scaled	Action	Adventure	Crime	Drama	History	Mystery	Sci-Fi
0	Inception	0.818182	1	1	0	0	0	0	1
1	Interstellar	0.636364	0	1	0	1	0	0	1
2	The Dark Knight	1.000000	1	0	1	1	0	0	0
4	Dunkirk	0.000000	1	0	0	1	1	0	0
5	The Prestige	0.545455	0	0	0	1	0	1	1

Cleaned dataset saved as 'cleaned_movie_ratings.csv'